PDM Technical Overview

Progressive Depletion Minting (PDM): Mechanism, Guarantees, and Architecture

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PDM is a non-discretionary supply control system. This document describes how it works, what it guarantees, and how it sits within the wider specification stack.

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1. About This Document

This Technical Overview sits between the Institutional Brief and the PDM Personal Edition Whitepaper, which contains the formal specification of the control law, proofs of the five structural guarantees, and empirical verification in Appendix C across seven test regimes. The companion Simulator Supporting Documentation extends this with a 625-point parameter stability sweep. The Overview is written for two audiences in parallel: senior policy advisors and chief economists who need to understand the mechanism in depth without working through formal proofs, and technical reviewers who will go on to read the Whitepaper but want a structured orientation first.

The document is self-contained. Reading it does not require prior familiarity with PDM, control theory, or any specific economic or computer-science background. Where mathematical notation appears, it is introduced on the page rather than assumed. Where the document refers to formal results, they are stated and explained, not reproduced in proof form. The full proofs are in the Personal Edition Whitepaper.

By the end, a reader should be able to describe the mechanism accurately, recognise its scope and its limits, and decide whether their institution would benefit from a deeper technical review.

2. The Problem PDM Addresses

Every system that manages a bounded resource pool faces the same governance question: when, and by how much, should the pool be expanded? Money supply, energy capacity, carbon allowances, compute allocation, regulated quotas, network bandwidth - the question is the same. The pool is finite, the demands on it are not, and someone has to decide when to add more.

In almost every existing system, that decision is discretionary. A committee meets. An executive signs off. A market participant reacts. The decision is made by humans applying judgement to inputs that are themselves contested. Three problems follow.

Discretion is slow

The decision arrives later than the condition that warranted it. By the time the decision is made, the underlying state has often moved further. Discretionary expansion lags the system it is meant to govern.

Discretion is opaque

Outside observers cannot verify why a particular decision was made. They can read the minutes, the press release, the explanatory speech. But the underlying logic that mapped state to action is not auditable. Institutions can act in good faith and still face credibility costs because the logic is invisible.

Discretion accumulates

Each individual expansion decision may be defensible in isolation. The cumulative effect over decades is what produces the credibility deficits institutions now face. There is no structural mechanism preventing political incentives from biasing expansion in one direction over time.

PDM addresses these three problems directly. It removes discretion from the expansion decision. It makes the logic that maps state to action fully auditable. And it builds in a structural resistance that makes expansion progressively harder over time, by mathematics rather than by judgement.

3. The Mechanism in Detail

PDM is a state transition function. On every operating cycle, the mechanism takes three inputs (current supply, measured obligations, measured activity) and produces a new supply value. The transition has three components, applied in sequence.

3.1 Continuous contraction (the burn)

On every cycle, an amount is removed from supply proportional to measured activity. The contraction rate is modulated by how far the system's velocity (activity divided by supply) deviates from a target ratio anchored at $\varphi \approx 0.618$. When the system is operating near the target, contraction is moderate. When it is far from the target, contraction adjusts.

The expression that governs contraction is:

$$burn = B \cdot \max(0, 1 - k \cdot (V/S - \varphi)) \cdot V$$

where B is a base contraction coefficient, k is a velocity sensitivity parameter, V is measured activity, and S is current supply. The terms B and k are operating parameters set by the adopting institution. φ is fixed at the golden ratio anchor.

The contraction component runs every cycle, regardless of system state. It is the default. It does not require any condition to fire.

3.2 Conditional expansion (the mint)

After contraction, the mechanism evaluates the supply-to-obligation ratio L = S / O, where O is measured obligations. This ratio is compared to a stability band defined by a lower threshold bL and an upper threshold bH, both close to φ. If L is within or above this band, no expansion occurs. Mint is exactly zero.

If L has fallen below bL, the mechanism computes a restoration amount intended to bring the system back toward the target ratio:

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mint_raw = max(0, φ · O − S)
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The trigger is a measured condition. There is no judgement involved. The mechanism does not consider whether expansion would be politically convenient, fiscally desirable, or strategically aligned. It evaluates the ratio. If the ratio is low enough, expansion is calculated. Otherwise, expansion is zero.

3.3 Progressive resistance (the damping)

When mint_raw is non-zero, the actual amount added to supply is reduced by a damping factor λ that depends on how full the system already is:

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λ = φ^(S / M)
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where M is the system's hard capacity ceiling. The notation x^y means x to the power y. When supply is small relative to M, λ is close to 1 and the full mint_raw is delivered. When supply approaches M, λ approaches zero and the mint delivers progressively less.

This is the property that gives PDM its name. The system becomes harder to expand the more it has already been expanded. There is no inflection point, no threshold, no committee decision to slow expansion. The resistance is built into the function. At 50% of capacity, λ ≈ 0.79: roughly four-fifths of the calculated mint is delivered. At 90% of capacity, λ ≈ 0.62. At 99% of capacity, λ ≈ 0.49 of an already smaller mint, of which only a fraction will fit before hitting the ceiling.

3.4 Capacity enforcement

Finally, the new supply value is clamped to the ceiling. If the damped mint would push supply above M, the actual mint is reduced so that the new supply equals exactly M, never more. The hard cap is unconditional. There is no override that allows supply to exceed M.

The complete state transition function, applied on every cycle, is:

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S_new = min(M, max(0, S − burn) + λ · mint_raw)
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with the convention that mint_raw is computed against (S − burn), and that mint_raw is zero unless the post-burn ratio is below the stability band.

4. The Five Formal Guarantees

Five structural properties hold for the state transition function that defines PDM. They hold unconditionally, regardless of who operates the mechanism, what parameters they choose, or what telemetry they feed into it. Each is stated below in plain language, with the underlying formal treatment held in the Whitepaper.

Guarantee	What it means
Non-negativity	Supply can never go below zero, regardless of telemetry inputs or parameter choices.
Capacity boundedness	Supply can never exceed the hard ceiling M. There is no override or exception that bypasses this.
Conditional minting	Expansion occurs only when the supply-to-obligation ratio L falls below the lower stability band. If L is healthy, expansion is exactly zero.
Progressive resistance	The damping factor λ = φ^(S/M) decreases monotonically as supply approaches the ceiling. Each successive expansion delivers less than the one before.
Deterministic auditability	Every operating cycle is recorded as a complete trace and chained via SHA-256. The full history is tamper-evident and independently verifiable.

The PDM Personal Edition Whitepaper provides the formal specification of the control law, including explicit statements and proofs of the structural invariants of the state transition function. These proofs establish the properties listed above - non-negativity, capacity boundedness, conditional minting, and progressive resistance - directly from the construction of the update function. They do not claim global optimality, full system stability under all possible telemetry conditions, or integrity of input data, which are addressed separately through empirical validation and the telemetry framework. Appendix C of the Whitepaper presents reproducible simulation results verifying these properties against the reference implementation.

Stability beyond the structural theorems is empirical. The Whitepaper's Appendix C presents seven test regimes empirically verifying all five guarantees against the reference implementation, including equilibrium stability, demand shock response, progressive resistance across a supply sweep, and cap enforcement. The companion Simulator Supporting

Documentation extends this with a 625-point parameter stability sweep across the burn base coefficient and velocity sensitivity, finding that the contraction coefficient remains positive across all tested configurations. Both analyses are reproducible by any reviewer running the simulator with the same parameters.

5. Domain Agnosticism

PDM operates on three abstract quantities: supply (S), obligations (O), and activity (V). The control law does not care what these quantities physically represent. The same state transition function applies whether the system being governed is a money supply, an energy grid, a carbon credit registry, a compute pool, or a regulated quota.

This is not aspiration. It is a property of the function itself. The mathematics is identical across sectors. What changes from sector to sector is the mapping from real-world quantities to S, O, and V, and the choice of operating parameters appropriate to that sector.

Domain	Supply (S)	Obligations (O)	Activity (V)
Money creation	Money supply	Credit demand	Settlements
Energy grid	MW capacity	Grid demand	Energy delivered
Carbon credits	Credits issued	Compliance obligations	Credits retired
Bandwidth	Capacity	Traffic demand	Throughput
Compute	Compute credits	Job queue	Jobs processed

The cross-sector applicability is what makes PDM a platform-level mechanism rather than a sector-specific innovation. A central bank deploying PDM and an energy regulator deploying PDM are using the same mathematics. The certification process, the audit chain format, and the structural guarantees are common across deployments. Only the parameter choices differ.

6. The Telemetry Layer

The control law's guarantees hold given valid telemetry inputs. They make no claims about the authenticity of those inputs. If the obligations and activity values fed into the mechanism are manipulated, then a formally verified control law will still produce expansion in response to false signals. The mechanism degrades to garbage in, garbage out.

This is not a flaw in the control law. It is a deliberate separation of concerns. A control law and an oracle are distinct engineering problems. Conflating them produces systems that are either

mathematically elegant but operationally fragile, or operationally robust but mathematically opaque. Separating them makes both tractable.

PDM's companion specification, O_n as Auditable Ledger, addresses the telemetry problem in detail. The summary version: telemetry integrity is achieved through a fixed ontology of admissible obligation types, a dual-key validation requirement on every entry, automatic temporal expiry, netting and deduplication, cryptographic attestation, and structural separation between the authority that benefits from issuance and the authority that validates the obligations driving it.

The combined effect of the control law and the telemetry framework is that manipulation of the system cannot be absorbed into the issuance rule. Any deviation from the rule must occur as an explicit, named, time-limited override that is cryptographically logged and publicly verifiable. The mechanism does not eliminate the possibility of override. It eliminates the possibility of hidden override.

7. Auditability

Every operating cycle produces a trace: a complete record of the inputs received, the intermediate values computed, and the new supply value produced. Each trace is serialised, hashed using SHA-256, and chained to its predecessor. The chain is append-only and tamper-evident. Any modification to a historical trace invalidates every subsequent hash and is therefore detectable.

The hash chain is not an optional feature. It is built into the state transition function itself. It cannot be turned off without modifying the mechanism, and a modified mechanism is not PDM.

The audit chain provides three operational properties. First, an external auditor can verify the integrity of the full operating history without privileged access to the underlying system. Second, the provenance of every supply value is reconstructible from the chain. Third, any party with access to the public chain head can independently confirm that the operating record has not been altered.

Auditability extends through the telemetry layer. The obligation register that feeds O_n to the control law maintains its own hash chain, distinct from the control law's chain but linked to it at each cycle. The two chains together form a complete audit surface from raw telemetry submission through to control law output.

8. Comparison with Existing Approaches

PDM is best understood by comparison with the three classes of existing approach to bounded supply control: commodity-backed standards, discretionary fiat regimes, and fixed-schedule

digital systems. Each prior approach has at least one of the four desirable properties. PDM is the first mechanism to combine all four.

Property	Commodity standard	Discretionary fiat	Fixed-schedule digital	PDM
Bounded supply	Yes (physical)	No	Yes	Yes
Responds to obligations	No	Yes (discretionary)	No	Yes (deterministic)
Progressive resistance	No	No	No	Yes
Auditable history	Partial	No	Yes	Yes
Discretion-free	Yes	No	Yes	Yes

Commodity standards (gold and other physical bases) are bounded and discretion-free, but unresponsive. They cannot expand to meet legitimate growth in obligations, producing deflationary crises. They also rely on physical custody, which introduces a separate trust layer.

Discretionary fiat is responsive, but unbounded and discretionary. It produces the three failure modes outlined in Section 2: lag, opacity, and accumulation. Every existing major currency operates in this category.

Fixed-schedule digital systems (Bitcoin and its derivatives) are bounded, auditable, and discretion-free, but they are not responsive to system state. Their issuance schedule is pre-committed at design time and cannot adapt to actual obligations. They are deflationary by construction.

PDM is the first design to combine boundedness, responsiveness, progressive resistance, auditability, and freedom from discretion in a single mechanism. The cell highlighted as 'Yes' under PDM in every row is the structural innovation.

9. Operating Parameters and Institutional Sovereignty

PDM's mathematics is universal. Its parameters are not. Each adopting institution chooses the parameters that govern its own deployment, within the constraints required for the mathematics to remain coherent.

The principal parameters are the maximum capacity M, the starting supply S_0, the lower and upper stability band thresholds bL and bH, the base contraction coefficient B, and the velocity sensitivity k. These choices determine how the mechanism behaves within the institution's specific context. A central bank operating under one mandate will choose different parameters

than one operating under another. An energy regulator in a peak-load environment will choose different parameters than one in a baseload environment.

This is the locus of institutional sovereignty within the framework. The mechanism does not tell the institution what its capacity ceiling should be, what its target ratio means in policy terms, or how aggressively the system should contract. The institution sets those values. Once set, the mechanism executes them deterministically and transparently.

This separation between parameters (chosen by the institution) and mathematics (fixed by the specification) is what allows PDM to operate across jurisdictions and sectors without imposing substantive policy. The mechanism is an enforcement layer for whatever rules the institution chooses to bind itself to.

10. What PDM Does Not Do

Honest demarcation of the mechanism's limits is part of what makes it credible. The following are explicitly out of scope.

- PDM does not specify who receives newly minted supply. Distribution policy is a separate concern of the deploying institution. The Cantillon problem (the question of which actors benefit first from new issuance) is not solved by PDM alone and must be addressed by a complementary distribution policy layer.
- PDM does not solve the oracle problem on its own. The control law's guarantees apply to the function as defined; the integrity of the inputs is the subject of the telemetry companion specification. Without the telemetry framework, a deployment is exposed to input manipulation.
- PDM does not protect against a sovereign or institutional actor that refuses to be bound by its own rules. No technical or mathematical system can substitute for rule-of-law enforcement. The framework's guarantees apply within jurisdictions that operate under enforceable separation of powers and judicial review.
- PDM does not claim sector-specific performance. Empirical claims about how a deployment will behave in a particular sector require real-world deployment data. Sector claims in the absence of such data are flagged as unproven throughout the specification stack.
- PDM does not eliminate uncertainty. It removes one specific source of systemic failure: uncontrolled, discretionary expansion. Other forms of uncertainty persist and are properly the domain of policy, not mechanism.

11. The Full Specification Stack

This Technical Overview is one document in a coordinated specification stack. Readers seeking deeper engagement should consult the documents below in the order most appropriate to their interest.

Document	Purpose
PDM Personal Edition Whitepaper	Full mathematical specification: state transition function, formal proofs of four theorems (Theorem 3 presented as two lemmas) yielding five structural guarantees, control-theoretic interpretation, simulation protocol, and Appendix C with seven test regimes empirically verifying all guarantees.
Reference Simulator and Supporting Documentation (Personal Edition)	Browser-based implementation reproducing the exact control law from the Go reference codebase, with built-in stability analysis tools, regime sequencer, and a 625-point parameter sweep. Accompanied by Supporting Documentation describing simulator architecture, telemetry profiles, dual-candidate Lyapunov analysis, and reproducibility procedure.
Equilibrium Demonstrator	Plain-language companion to the simulator for non-technical audiences. Live charts, audit chain verification, and sector context selector.
Telemetry Companion Paper (O_n as Auditable Ledger)	Specification for the integrity of inputs to the control law: ontology, dual-key validation, expiry, netting, attestation, separation of powers, constrained override.
Licensing and Certification Framework	Three-layer governance model: open study, commercial licensing, and independent certification through MannCert.
Institutional Brief	Reference document for ministers, central banks, and senior policy leaders. High-level overview of the framework and engagement pathway.
Institutional Engagement Fee Schedule	Published fee structure for the three-stage advisory engagement pathway.
Technical Overview (this document)	Mid-layer technical bridge between the Institutional Brief and the Personal Edition Whitepaper. Suitable for senior policy advisors, chief economists, and technical reviewers.

A reader from a policy background may wish to follow this document with the Telemetry Companion Paper, then the Licensing and Certification Framework. A reader from a technical background may wish to proceed directly to the Personal Edition Whitepaper and the reference simulator. Both routes converge on the same complete picture; the order of approach is a matter of preference.

12. Status and Intellectual Property

Progressive Depletion Minting is covered by a pending UK patent (UKIPO GB2513172.3, filed August 2025) held by Mann Mechanics Ltd. The patent is at pending status; grant has not yet been issued. The Personal Edition reference simulator is published openly for non-commercial study and verification.

Commercial deployment of PDM in any sector requires a commercial licence from Mann Mechanics Ltd. Independent conformance certification, where required by the deployment's regulatory or public-trust standing, is provided by MannCert Ltd. Mann Mechanics and MannCert are separately constituted entities with distinct functions; their structural separation is a deliberate governance feature.

The full licensing structure is specified in the PDM Licensing and Certification Framework document. Inquiries regarding commercial licence terms are directed to Mann Mechanics. Inquiries regarding institutional certification are directed to MannCert.

13. Closing

PDM is a mechanism, not a doctrine. It is a state transition function that takes three inputs and produces a new supply value, with five mathematically guaranteed properties and a complete audit chain. The mechanism is what it is. It does not require belief, endorsement, or political alignment to function.

The case for adopting it rests on whether an institution's circumstances are improved by replacing discretionary expansion decisions with a deterministic, auditable, progressively resistant alternative. That judgement belongs to the institution. The framework provides the mechanism, the specifications, the certification structure, and the engagement pathway. What it cannot provide is the institutional decision to engage.

This document is offered as a structured basis for that decision. The supporting specifications are available for any institution that wishes to evaluate the mechanism in detail.

Disclaimer

This document is a technical overview of the PDM mechanism and the supporting specification stack. It does not constitute legal, financial, regulatory, or monetary policy advice. It is not an offer to license or a binding commitment. Sector-specific performance claims are unproven in the absence of real-world deployment data. Institutions remain solely responsible for their own evaluation, decision-making, and policy outcomes.

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